Stair Stringers and Treads

Building Code Requirements

Stair Stringer to Rim Joist

- Stairs must be able to resist 40 lb. per square foot of tread area in one and two family dwellings. All other stairs under IBC require a design to 1.5 times the live load for the area served, not to exceed 100 psf.

IRC 2018/2021/2024 Table R301.5
IBC 2018/2021/2024 Table 1607.1

Stair Tread to Stringer

- Individual stair treads shall be able to resist a 300 lb. concentrated load acting over an area of 2" x 2".

IRC 2018/2021/2024 Table R301.5
IBC 2018 Table 1607.1
IBC 2021/2024 Table 1607.1
Section 1607.21

Guard Opening Limitations

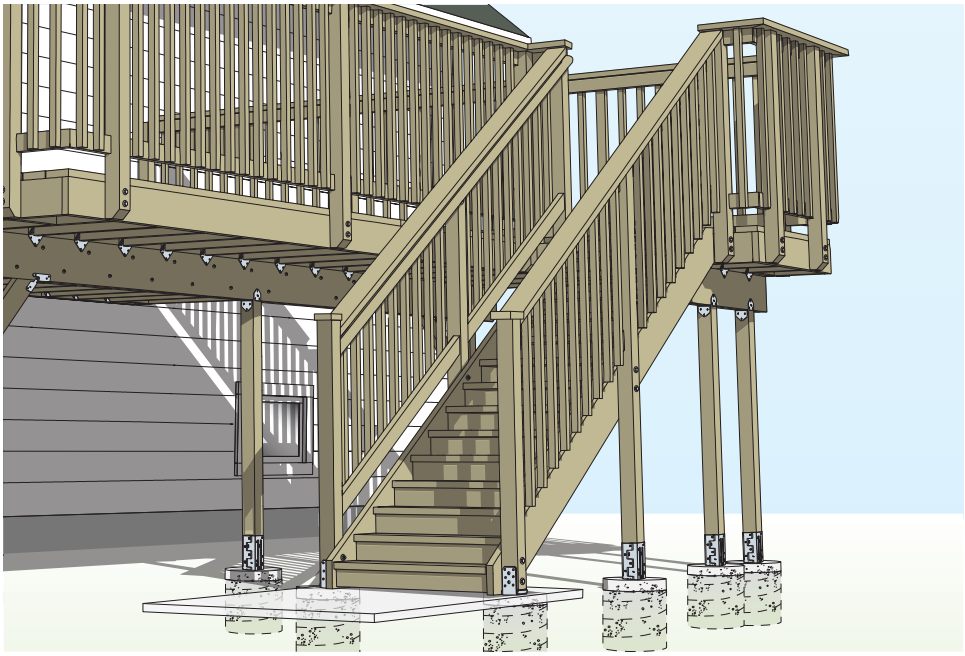
- The triangular opening formed by the riser, tread and bottom rail of a guard at the open side of a stairway shall be of such a size that a sphere 6" in diameter cannot pass through.

IRC 2018/2021 Section R312.1.3
IRC 2024 Section R321.1.3
IBC 2018/2021/2024
Section 1015.4

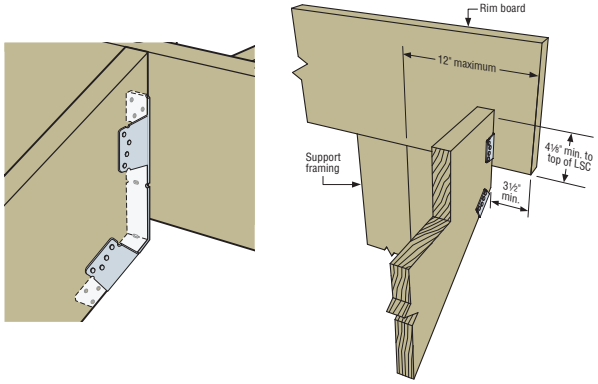
Additional Exception: Openings for required guards on the open side of stairs shall not allow a sphere 4⅜" in diameter to pass through. For IBC, exception is only valid for individual dwelling units in Group R-2 and R-3 occupancies.

IRC 2018/2021/2024
Section R312.1.3
IRC 2024 Section R321.1.3
ICB 2018/2021/2024
Section 1015.4

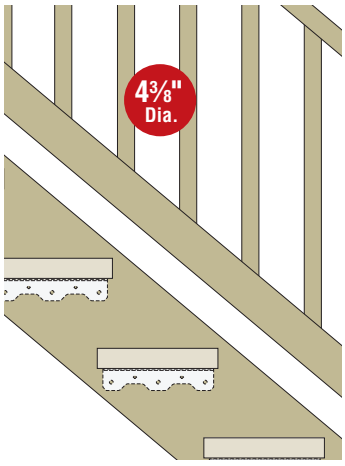
Stair stringers must be properly connected to the deck, and treads properly connected to the stringers, in order to resist loads. In addition, code requirements regarding openings between stair treads and stair railing must also be met.



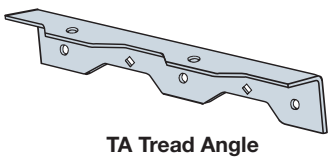
Simpson Strong-Tie® Solutions



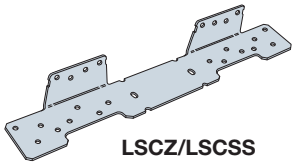
LSC Adjustable/LSC Cantilever Stair Stringer Connector
Field slopeable to almost any angle, the LSC is ideal for attaching the staircase stringer to the deck. Suitable for solid and notched stringers. Available in ZMAX® and Type 316 stainless steel. For information regarding notched stringer attachment to full-width stringers, see L-F-DCA6SCREW.



Codes require that a 4⅜" sphere must not be able to pass between any guard opening at stairs.



TA Tread Angle
Provides a positive connection, eliminating the need to notch the stringer. A full-width stringer is stronger and makes it easier to meet maximum guardrail opening requirements of the codes. ZMAX coating or stainless steel required.



LSCZ/LSCSS



Strong-Drive® SDS Heavy-Duty Connector Screw
(Included in TA9ZKT and TA10ZKT kits)



Strong-Drive SD Connector Screw
(for use with LSCS/LSCSS)

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